

EDUCATION

Guru Nanak Dev University, Amritsar (July 2024 - July 2028) Bachelor of Technology CSE - 8.0/10	Spring Dale Senior School, Amritsar 10th CBSE - 95% 12th CBSE - 87.5%
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PROJECTS

AI-Driven Medical Report Analysis & Appointment System - [Github](#)

- AI Pipeline:** Architected an end-to-end AI pipeline using OCR and GROK AI to automate symptom extraction, generating high-precision medical summaries with verified accuracy.
- Recommendation:**Designed a doctor recommendation engine leveraging contextual symptom matching and weighted scoring logic to route patients to relevant specialists
- API & Security:** Deployed enterprise-grade RESTful APIs via Django Ninja and JWT-based RBAC, securing complex appointment workflows for Admin, Doctor, and Patient user roles.
- Performance:** Maximized backend performance and reliability by integrating Redis caching and Celery task queues, slashing request latency by 60% and eliminating redundant blocking operations.

Tech Stack: Django, Django-Ninja, GROK API, React, Langchain, PostgreSQL, Redis, Celery, OCR

Automated YouTube Shorts Generation Pipeline - [Github](#)

- Scene Detection:** Architected an automated vision pipeline using OpenCV to analyze scene dynamics and activity spikes, programmatically distilling long-form videos into high-engagement vertical Shorts.
- Subtitle Engine:** Engineered a high-fidelity subtitle engine leveraging OpenAI Whisper and ASS styling to automate localized, professional captioning with dynamic typography and precise temporal alignment.
- FFmpeg Optimization:** Optimized low-latency media workflows using FFmpeg for parallelized trimming, resizing, and spatial filtering, ensuring high-throughput processing for complex video transformation tasks.
- Frontend UX:** Developed a high-performance React interface featuring real-time rendering of subtitle styles and state-driven previews, significantly enhancing the end-to-end user content creation workflow.

Tech Stack: Django, Django-Ninja, React, JavaScript, TailwindCSS, PostgreSQL, OpenCV, Ffmpeg, Whisper

Location-Based Farmers E-Commerce Platform - [Github](#)

- Platform Vision:** Engineered a hyper-local e-commerce ecosystem to decentralize agricultural supply chains, leveraging geospatial logic to connect consumers directly with regional producers.
- Backend Architecture:** Architected high-performance RESTful API's using Django Ninja, optimizing API endpoints for scalable product indexing and multi-role user state management.
- Geospatial Integration:** Integrated Google Maps API to implement discovery of sellers within a 50km radius based on user coordinates.
- Data Integrity:** Designed a normalized PostgreSQL schema to maintain strict ACID compliance for transactional data, ensuring reliable inventory tracking and secure user account management.

Tech Stack: Django, Django-Ninja, JavaScript, Bootstrap, PostgreSQL, Google-Maps API , render

SKILLS

Languages: Python, JavaScript, C++, HTML/CSS
Frameworks: Django, Django-Ninja, React, Next Js, LangChain
Technologies: REST APIs, OpenAI API, Celery, JWT, Github, Docker, Google ADK
Databases: PostgreSQL, SQLite , Redis, Supabase

ACHIEVEMENTS

HACKERWRATH 2.0 (2nd Place): "Developed a hybrid AI-Human chatbot using Retrieval-Augmented Generation (RAG) and real-time sentiment analysis, featuring a human-in-the-loop oversight dashboard."
Node Hash (1st Place): "Architected a full-stack multimedia productivity platform for automated task queuing and dynamic monthly report generation."